

Description of interface commands

Valid for oscilloscopes:

HM305-2, HM404, HM407, HM1004-2, HM1505-2, HM1507, HM1507-2

RS-232 cable

A 9 conductor serial interface cable (1:1 connection, i.e. without crossed pin connections) will be required to connect the oscilloscope to an external instrument.

Setting the baud rate

The RS 232 interface must be initialized before use. This is effected by the first transmission of SPACE CR (20 hex., 0D hex.) after POWER UP (*switching on*). This will automatically set the baud rate. The following baud rates will be recognized:

110 Baud	4800 Baud
150 Baud	9600 Baud
300 Baud	19200 Baud
600 Baud	38400 Baud ^(1.19)
1200 Baud	56700 Baud ^(1.19)
2400 Baud	115200 Baud ^(1.19)

Data transmission format:

No parity, data length 8 bits,
2 stop bits, RTS/CTS handshake.

If the PC has a COM interface with FIFO buffer, then the maximum depth of the send buffer must be set to 8 bytes.

The baud rate set will remain operative until POWER DOWN (*switch off*) or until the remote mode is disabled with the command 'RMO', or until the pushbutton LOCAL (AUTOSSET button) is activated, assuming this is not inhibited (LK=0).

Once the remote mode is disabled, the data transmission can be only restored by again sending SPACE CR.

If the oscilloscope does not recognize SPACE CR as first characters, or if Low-level exists on the RTS line longer than for about 2 seconds, then the oscilloscope exits the remote mode and sets the TxD Low for about 0.3s and thus generates a frame error.

The oscilloscope answers with a RETURNCODE (0 CR LF) if it has recognized SPACE CR and has set the baud rate.

The scope sends 'ESC RMLK=0' (ESC=1B hex.) and exits the Remote state if the button LOCAL is activated in Remote-ON state. The time between the reception of the RETURNCODE Remote-OFF and Remote-ON must be at least

$$t_{\min} = 2 \cdot (1/\text{Baudrate}) + 60\mu\text{s}$$

Data transmission

After successfully setting the baud rate, the oscilloscope is in Remote state and is ready to accept commands. The commands can be in capital or lower case. The commands can be divided into two basic groups:

Interrogation of parameters

This group of commands is distinguished by a question mark at the end of the command. When such a command is transmitted to the scope, it answers by repeating the syntax followed by a colon and the parameters asked for. These can be binary or ASCII data depending on the command. The number of data bytes to be received is dependent on the command and can be seen from the command description.

Example: Command to the scope: VERS?
 Answer: VERS:FC1.01 DG1.02

Set parameter

The parameters of the scope can be influenced with this command. Here commands with and without parameter can be differentiated.

Example: Command to the scope: LK=1
 Answer: RETURNCODE

Example: Command to the scope: RES
 Answer: RETURNCODE

All commands are answered either with parameters or with a RETURNCODE in ASCII format (see description of commands). One must wait for all parameters or RETURNCODE for the previous command before a new command can be sent to the scope.

WORD-parameters require first the low byte and then the high byte. The setting of the scope is effected over the instrument data field (DeviceDataField DDF) as *binary array*. Each byte of this data field can also be accessed by individual commands.

The following tables show the build up of the instrument data field and the corresponding individual commands.

Definition of characters for the commands

Interrogation	?	Interrogates for parameters
Assign	=	Set parameter
State	:	Gives current parameter
Binary data	b	Data field is binary data of 1 byte
ASCII data	a	Data field is ASCII data of 1 Byte
WORD (2 bytes)	w	Data field consists of 2 bytes (low and high bytes)
ASCII data	ARRAY	Data field is ASCII data
Binary data	array	Data field is binary data
Terminating character	CR LF	Carriage return and/or Line feed
RETURNCODE	R	ASCII parameter
Parameter	x	x stands for A or B
Parameter	z	z stands for 1 or 2

Table of commands:

Command: PC -> Scope	Acknowledgment Scope -> PC	Description
AUTOSET	(R CR LF)	AUTO SET function will be carried out
AVRNM? ⁽³⁾	AVRNM:(b)	AVERAGE NUMBER OF ACQUISITIONS Delivers the number of acquisitions carried out for averaging b = 01 hex.: 2 ¹ = 2 acquisitions b = 02 hex.: 2 ² = 4 acquisitions ... b = 09 hex.: 2 ⁹ = 512 acquisitions
AVRNM=(b) ⁽³⁾	(R CR LF)	AVERAGE NUMBER OF ACQUISITIONS Sets the number of acquisitions to be carried out for averaging
AVRNMSW? ⁽³⁾⁽⁵⁾	AVRQTSW:(a)	AVERAGE NUMBER OF ACQUISITIONS SWITCH Delivers the function of the Store Up/Down buttons a = 0: Store Up/Down buttons change Store Mode a = 1: Store Up/Down buttons change Average Number of Acquisitions
AVRNMSW=(a) ⁽³⁾⁽⁵⁾	(R CR LF)	AVERAGE NUMBER OF ACQUISITIONS Sets Average Number of Acquisition Switch
BELL=(a)	(R CR LF)	Tone output a = 0: Buttons OK tone a = 1: Buttons ERROR tone a = 2: ERROR (longer tone) a = 3: 2 short tones a = 4: 3 short tones a = 5: 6 short tones
CH<z>?	CH<z>:(b)	Delivers CH1/2 settings see table of instrument fields DDF
CH<z>=(b)	(R CR LF)	Sets CH1/2 settings see table of instrument fields DDF
CH<z>VAR?	CH<z>VAR:(b)	Delivers CH1/2 VARI-GAIN setting b = FF hex.: CH<z> (1 or 2) calibrated
CH<z>VAR=(b)	(R CR LF)	Sets CH1/2 VARI-GAIN setting
CTRLBP?	CTRLBP:(a)	Delivers CONTROL BEEP setting a = 0: Off (key activation without control tone) a = 1: On
CTRLBP=(a)	(R CR LF)	Sets CONTROL BEEP setting
DDF?	DDF:(array)	Delivers DEVICE DATA FIELD see table of instrument fields (4 bytes of commands + 14 bytes of parameters)
DDF=(array)	(R CR LF)	Sets new DEVICE DATA FIELD see table of instrument fields DDF

Command: PC -> Scope	Acknowledgment Scope -> PC	Description
DDF1?	DDF1:(array)	Delivers DEVICE DATA FIELD1 see table of instrument fields DDF1 (5 bytes of commands + 16 bytes of parameters.)
DDF1=(array)	(R CR LF)	Sets new DEVICE DATA FIELD1 see table of instrument fields DDF1
DELPOS?	DELPOS:(w)	Delivers DELAY POS. (12 Bit) w = 000 hex.: Shortest DELAY time
DELPOS=(w)	(R CR LF)	Sets DELAY POS. (12 bit)
ERRBP?	ERRBP:(a)	Delivers ERROR BEEP setting a = 0: Off (no error tone output) a = 1: On (with error tone output)
ERRBP=(a)	(R CR LF)	Sets ERROR BEEP
ERRMSG?	ERRMSG:(a)	Delivers ERROR MESSAGE setting a = 0: Off (Error messages are only sent to the interface) a = 1: On (Error messages are displayed with Read out and also sent to the interface)
ERRMSG=(a)	(R CR LF)	Sets ERROR MESSAGE setting
FCCMD? ^(1,1)	FCCMD:(a)	FRONT CONTROLLER COMMAND Informs whether the scope has been manually operated. Will be reset after each interrogation.. a = 0: Scope has not been manually operated. a = 1: Scope has been operated manually (in the mean time).
HLD<z>POS? ⁽³⁾	HLD<z>POS:(b)	Delivers HOLD 1/2 POSITION Y-offset carried out after HOLD referred to Store position. b = 00 hex.: Position shifted maximum upwards.
HLD<z>POS=(b) ⁽³⁾	(R CR LF)	Sets HOLD 1/2 POSITION
HOLDOFF?	HOLDOFF:(b)	Delivers HOLD OFF value b = 00 hex.: Shortest Hold off time
HOLDOFF=(b)	(R CR LF)	Sets HOLD OFF value
HORMODE?	HORMODE:(b)	Delivers HORIZONTAL MODE setting see table of instrument fields DDF
HORMODE=(b)	(R CR LF)	Sets HORIZONTAL MODE setting see table of instrument fields DDF
HLDWFM? ⁽³⁾	HOLDWFM:(a)	HOLD WAVE FORM Delivers HOLD function setting in Store mode a = 0: Off (HOLD switched off) a = 1: On (HOLD switched on)
HLDWFM=(a) ⁽³⁾	(R CR LF)	HOLD WAVE FORM Sets HOLD setting

Command: PC -> Scope	Acknowledgment Scope -> PC	Description
ID?	ID: (ARRAY)(CR LF)	IDENTIFICATION (Hardware) Delivers instrument name and hardware identification bytes (3 bytes of commands + 27 bytes of parameters)
INT<x>?	INT<x>:(b)	Delivers INTENS A/B ⁽⁹⁾ value b = 00 hex.: Trace blank timebase <x> (A or B ⁽⁹⁾)
INT<x>=(b)	(R CR LF)	Sets INTENS A/B ⁽⁹⁾ value
INTRO?	INTRO:(b)	Delivers INTENS READ OUT value b = 00 hex.: Read out blanked
INTRO=(b)	(R CR LF)	Sets INTENS READ OUT value
LK?	LK:(a)	Delivers the function of LOCAL LOCK OUT key (AUTO SET) a = 0: Locked a = 1: Free (The Remote mode will be exited on activation of the AUTOSET button and 'RMLK=0' will be sent to the interface)
LK=(a)	[R](CR LF)	Setting the LOCAL LOCK OUT function
PSINT?	PSINT:(a)	Delivers PULSE SWITCH INTENS function of the rotary control: INTENS a = 0: INT A (Intens. trace A) a = 1: INT RO (Intens. Read out) a = 2: INT B (Intens. trace B) ⁽⁹⁾
PSINT=(a)	(R CR LF)	Sets PULSE SWITCH INTENS function of the rotary control: INTENS
PSY1POS?	PSY1POS:(a)	Delivers PULSE SWITCH Y 1 POSITION function of the rotary control: Y-POS. I a = 0: Y1 Position setting a = 3: Y Position setting of timebase B (Trace sep.) in alternating timebase mode ⁽⁹⁾
PSY1POS=(a)	(R CR LF)	Sets PULSE SWITCH Y 1 POSITION (function of the rotary control: Y-POS. I)
PSY2POS?	PSY2POS: (a CR LF)	Delivers PULSE SWITCH Y 2 POSITION function of the rotary control: Y-POS. II a = 0: Y2 Position setting
PSY2POS=(a)	(R CR LF)	Sets PULSE SWITCH Y 2 POSITION function of the rotary control: Y-POS. II
PSTB?	PSTB:(a)	Delivers PULSE SWITCH TB function of the rotary control: TIME/DIV. a = 0: COARSE (1-2-5 sequence) a = 1: FINE (variable, depending on the timebase mode A or B)
PSTB=(a)	(R CR LF)	Sets PULSE SWITCH TB function of the rotary control: TIME/DIV.

Command: PC -> Scope	Acknowledgment Scope -> PC	Description
PSCH<z>?	PSCH<z>:(a)	Delivers PULSE SWITCH CH1/2 function of the rotary control: VOLTS/DIV (CH I or CH II) a = 0: CH I or II COARSE (1-2-5 sequence) a = 1: CH I or II FINE (variable)
PSCH<z>=(a)	(R CR LF)	Sets PULSE SWITCH CH1/2 function of the rotary control: VOLTS/DIV (CH I or CH II)
QUICKST? ^(1.05)	QUICKST:(a)	Delivers QUICK START MODE a = 0: QUICK START off a = 1: QUICK START on
QUICKST=(a) ^(1.05)	(R CR LF)	Sets QUICK START MODE
RDWFM<z>: (w w) ⁽³⁾⁽⁷⁾ 1st WORD Offset hex. (2 Kbytes) 2nd WORD length hex. (2 Kbytes)	RDWFM<z>: (W W array) ⁽⁷⁾	READ WAVE FORM 1/2 Delivers signal data of channel <z> (1 or 2), from Offset address (first WORD) with the given length (second WORD) Offset + length max. 2 Kbytes Complete transmission from channel 1: Offset = 0, length = 2048 dec. (2 Kbytes) 'RDWFM1:00000008' (figures in hex.) Data from the right half of the signal of channel 2: Offset = 1024 dec., length = 1024 dec. 'RDWFM2:00040004' (figures in hex.)
READOUT?	READOUT:(a)	Delivers READ OUT setting a = 0: Read out off a = 1: Read out on
READOUT=(a)	(R CR LF)	Sets READ OUT setting
RDREF<z>: (w w) ⁽³⁾⁽⁷⁾ 1st WORD Offset hex. (2 Kbytes) 2nd WORD length hex. (2 Kbytes)	RDREF<z>: (W W array) ⁽⁷⁾	READ REFERENZ 1/2 Delivers Signal data from Reference store <z> (1 or 2), from Offset address (first WORD) with given length (second WORD) Offset + length max. 2 Kbytes See 'RDWFM<z>'
RECDF=(a)	(R CR LF)	RECALL DDF Reads instrument data field of „SAVE/ RECALL-Store“ (a = 1 ... 9) and sets the scope accordingly.
REF<z>POS? ⁽³⁾	REF<z>POS:(b)	Delivers REF 1/2 POSITION Y Position of the reference traces in Digital mode b = FF hex. REF<z> shifted max. upwards
REF<z>POS=(b) ⁽³⁾	(R CR LF)	Sets REF 1/2 POSITION

Command: PC -> Scope	Acknowledgment Scope -> PC	Description
REF<z>PRE? ⁽³⁾	REF<z>PRE: (array)	REFERENZ 1/2 PREAMBLE Sets the scope in the state in which the reference traces were stored. Besides, information will be sent to the PC. array: 1st WORD: Memory address at the time of triggering 2nd WORD: X resolution = 200 bit/DIV 3rd WORD: Y resolution = 25 bit/DIV 4th WORD: Y1 standardized position based on the value of WORD 3 (Integer value) 5th WORD: Y2 standardized position based on the value of WORD 3 (Integer value)
RES(CR LF) ⁽²⁾	(R CR LF)	Sets RESET function in SINGLE mode
RM0(CR LF)	(R CR LF)	REMOTE Exit remote mode
RREFPRE ⁽³⁾	(R CR LF)	RECALL REFERENZ PREAMBLE Sets the scope in the state in which the current acquisition was stored as reference. (HM1507/1507-2: The REF1 preamble will be loaded if both traces are switched on.)
RODDF?	RODDF:(array)	Delivers READ OUT DEVICE DATA FIELD (RO device data field) (6 bytes of commands + 10 bytes of parameters) See table read out data field RODDF
RODDF=(array)	(R CR LF)	Sets new READ OUT DEVICE DATA FIELD (RO device data field)
SAVEDF=(a)	(R CR LF)	SAVE DEVICE DATA FIELD Stores the current instrument settings in the „SAVE/RECALL-Store“ in memory position a (a = 1 ... 9)
SAVREF<z> (CR LF) ⁽³⁾	(R CR LF)	SAVE REFERENZ 1/2 In STORE MODE, stores the current signal data in reference store <z> (1 or 2)
STRMODE? ⁽³⁾	STRMODE:(b)	Delivers STORE MODE see table of instrument fields DDF
STRMODE=(b) ⁽³⁾	(R CR LF)	Sets STORE MODE see table of instrument fields DDF
TB<x>?	(R CR LF)	TIMEBASE A/B Sets timebase A/B setting see table of instrument fields DDF
TB<x>=(b)	TB<x>:(b)	TIMEBASE A/B Delivers timebase A/B setting see table of instrument fields DDF

Command: PC -> Scope	Acknowledgment Scope -> PC	Description
TB<x>VAR?	TB<x>VAR:(w)	Delivers TIMEBASE A/B VAR setting (10 bits) w = 000 hex.: TB<x> (A or B) calibrated w = 001 .. 3FF hex.: TB<x> uncalibrated
TB<x>VAR=(w)	(R CR LF)	Sets TIMEBASE A/B VAR setting (10 bit)
TRIG?	TRIG:(b)	Delivers TRIGGER parameter see table of instrument fields DDF
TRIG=(b)	(R CR LF)	Sets TRIGGER parameter see table of instrument fields DDF
TRSEP? ⁽⁹⁾	TRSEP:(b)	Delivers TRACE SEP. Y position of the timebase B referred to the Y position of the timebase A b = 00 hex.: B maximum above A b = FF hex.: B maximum below A b = 80 hex.: Y position of B = Y position of A
TRSEP=(b) ⁽⁹⁾	(R CR LF)	Sets TRACE SEP.
TRGSTA?	TRGSTA:(b) TRGSTA:(a) ^(1,17)	Delivers TRIGGERSTATUS a(or b) = 0: Instrument does not trigger a(or b) = 1: Instrument triggers a(or b) = 2: Instrument is in SINGLE RESET MODE or, acquisition not yet terminated.
TRGSTA(CR LF)	(R CR LF)	TRIGGERSTATUS Reset Automatic monoflop ($\approx 200\text{ms}$) Enables immediate interrogation with 'TRGSTA?'.
TRGVAL?	TRGVAL:(array)	TRIGGERVALUE Delivers trigger signal voltage value (measured at the trigger amplifier) array: 16 Bit INTEGER; 1st WORD: Positive peak value 2nd WORD: Negative peak value 3rd WORD: Peak to peak value 4th WORD: DC trigger working point (reference value) Weighting : - WORD 1-3 : ca. 20mV/LSB - WORD 4 : ca. 5mV/LSB and 250mV/DIV (2.00: 1st WORD: Positive peak value 2nd WORD: Negative peak value 3rd WORD: Mean arithmetic value 4th WORD: Reserved Weighting: 1000 Bit/DIV

Command: PC -> Scope	Acknowledgment Scope -> PC	Description
TRGLEV<x>?	TRGLEV<x>:(w)	Delivers TRIGGER-LEVEL A/B setting (10 bits) w = 3FF hex.: max. positive (right limit) w = 000 hex.: max. negative (left limit)
TRGLEV<x>=(w)	(R CR LF)	Sets TRIGGER-LEVEL A/B setting (10 bits)
VERMODE?	VERMODE:(b)	Delivers VERTICAL MODE (vertical setting) see table of instrument fields DDF
VERMODE=(b)	(R CR LF)	Sets VERTICAL MODE (vertical setting) see table of instrument fields DDF
VERS?	VERS:(ARRAY)	Delivers SOFTWAREVERSIONS 5 bytes of commands + 15 bytes of parameters
WFMPRE? ⁽³⁾	WFMPRE:(array)	WAVE FORM PREAMBLE Delivers Data for the stored traces array: 1st WORD: Memory address at the time of trigger 2nd WORD: X resolution = 200 bit/DIV 3rd WORD: Y resolution = 25 bit/DIV 4th WORD: Y1 position standardized on the value of WORD 3 (integer value) 5th WORD: Y2 position standardized on the value of WORD 3 (integer value)
WRREF<z>: ⁽³⁾⁽⁶⁾⁽⁷⁾ (w w array) 1st WORD Offset hex. (2 Kbytes) 2nd WORD length hex. (2 Kbytes)	(R CR LF)	WRITE REFERENZ 1/2 writes signal data into the reference store <z> (1 or 2), from offset address (first WORD) with given length (second WORD) Offset + length max. 2 Kbytes See 'RDWFM<z>'

Command: PC -> Scope	Acknowledgment Scope -> PC	Description
XPOS?	XPOS:(w)	Delivers X-POSITION setting (10 bits) w = 3FF hex.: „right limit“ w = 000 hex.: „left limit “ (1.1: 16 Bit INTEGER value in two's complement referred to graticule center (1000 bits/DIV) ⁽¹⁰⁾
XPOS=(w)	(R CR LF)	Sets X-POSITION setting (10 bits) (1.1: (16 Bit INTEGER)
Y<z>POS?	Y<z>POS:(w)	Y 1/2 POSITION Delivers CH<z> (1 or 2) position setting (10 bits) w = 3FF hex.: Y position maximum upwards w = 000 hex.: Y position maximum downwards (1.1: 16 Bit INTEGER value in twos complement referred to graticule center (1000 bits/DIV) ⁽¹⁰⁾
Y<z>POS=(w)	(R CR LF)	Y 1/2 POSITION Sets CH<z> (1 or 2) position setting (10 bits) (1.1: (16 Bit INTEGER)

All commands will be internally checked for any conflicts and protocolled in RETURNCODE.
The following RETURNCODES (ASCII characters) have been implemented:

- 0 = no error
- 1 = syntax error
- 2 = data error
- 3 = buffer overflow
- 4 = bad data set
- 5 = adjustment error
- 6 = timing error (internal data transmission FC /STORE)

Instrument data field (DDF)

	D7	D6	D5	D4	D3	D2	D1	D0
CH1	GND	AC	INV1	ON	VOLT/DIV - Counter 0-13 0000(1mV/DIV)...1101(20V/DIV)			
CH2	GND	AC	INV2	ON	VOLT/DIV - Counter 0-13			
VERMODE	Alt.-TRG	Probe CH1: ^(2.00) 0 = 1:1 1 = 10:1	0	CHOP	ADD	Probe CH2: ^(2.00) 0 = 1:1 1 = 10:1	TR-SOURCE 00 = CH1 01 = CH2 1x = EXT	
TBA	0	0	ANA SING	MODE Analog: Store: ⁽³⁾ Store XY: ⁽³⁾ Store ROLL: ⁽³⁾	TIME/DIV – Counter 00hex. - 15hex.{SEA:11hex} 50ns/DIV - 0,5s/DIV{SEA:20ms} 04{03 ⁽⁸⁾ }hex. - 1Chex 1µs{500ns ⁽⁸⁾ }/DIV - 100s/DIV 06{05 ⁽⁸⁾ }hex. - 1Chex 40MS{100MS ⁽⁸⁾ }/s - 2S/s 12hex. - 1Chex 50mS/s - 100S/s			
TBB	B +/- (1=neg. Trigger- edge)	B-TR	0	MODE Analog: Store: ⁽³⁾	TIME/DIV – Counter 00hex. - TBA(max.11hex.) 50ns - TBA(max.20ms/DIV) 04{03 ⁽⁸⁾ }hex. - TBA(max.11hex.) 1µs{500ns ⁽⁸⁾ }/DIV –TBA(max.20ms/DIV)			
HORMODE	CT (Compo- nents- tester)	XY	x10	STORE ⁽³⁾	0	TB-MODE HM1004-2/1505-2 HM305-2 HM1507/1507-2 HM404/407 000: TBA TBA 001: TBA + TBB reserv. 010: TBA alt.TBB SEARCH 011: TBB DELAY TRG		
TRIG	+/- (1=neg Trigger- edge)	0	P-P	NORM	0	Kopplung 0-7 HM1004-2/1505-2 HM305-2 HM1507/1507-2 HM404/407 000: AC AC 001: DC DC 010: HF HF 011: NR LF 100: LF TVLine 101: TVLine TVField 110: TVField LINE 111: LINE reserv.		
STRMODE ⁽³⁾	REF2	REF1	PRE TRIGGER 000 = -75% 001 = -50% 010 = -25% 011 = 0% 100 = 25% 101 = 50% 110 = 75% 111 = 100%			STOREMODE 000 = REF 001 = SGL 010 = ROL (TB=100s..50ms) 011 = ENV 100 = AVR		
CH2 VAR	8-BIT							
CH1 VAR	8-BIT							
TRSEP ⁽⁹⁾	8-BIT							
HOLD OFF	8-BIT							
INTENS A	8-BIT							
INT B ⁽⁹⁾	8-BIT							

Instrument data field 1 (DDF1)

	D15	D14	D13	D12	D11	D10	D9	D8	D7	D6	D5	D4	D3	D2	D1	D0
TR A LEV	0	0	0	0	0	0	X	X	X	X	X	X	X	X	X	X
TB A VAR	0	0	0	0	0	0	X	X	X	X	X	X	X	X	X	X
X POS	X (1.1)	X (1.1)	X (1.1)	X (1.1)	X (1.1)	X (1.1)	X	X	X	X	X	X	X	X	X	X
Y2 POS	X (1.1)	X (1.1)	X (1.1)	X (1.1)	X (1.1)	X (1.1)	X	X	X	X	X	X	X	X	X	X
Y1 POS	X (1.1)	X (1.1)	X (1.1)	X (1.1)	X (1.1)	X (1.1)	X	X	X	X	X	X	X	X	X	X
TR B LEV	0	0	0	0	0	0	X	X	X	X	X	X	X	X	X	X
TB B VAR	0	0	0	0	0	0	X	X	X	X	X	X	X	X	X	X
DEL POS	0	0	0	0	0	0	X	X	X	X	X	X	X	X	X	X

Read out data field (RODDF)

	D15	D14	D13	D12	D11	D10	D9	D8	D7	D6	D5	D4	D3	D2	D1	D0
CURSOR MODE	0	0= CH1 1= CH2	0	0=X 1=Y	1= Trk	0dt 1=f	0	1= Crs On	I	N	T	-	R	O	U	T
CURSOR X active	X (1.1)	X (1.1)	X (1.1)	X (1.1)	X (1.1)	X (1.1)	X	X	X	X	X	X	X	X	X	X
CURSOR X passive	X (1.1)	X (1.1)	X (1.1)	X (1.1)	X (1.1)	X (1.1)	X	X	X	X	X	X	X	X	X	X
CURSOR Y active	X (1.1)	X (1.1)	X (1.1)	X (1.1)	X (1.1)	X (1.1)	X	X	X	X	X	X	X	X	X	X
CURSOR Y passive	X (1.1)	X (1.1)	X (1.1)	X (1.1)	X (1.1)	X (1.1)	X	X	X	X	X	X	X	X	X	X

(1.05 From FC-Version 1.05

(1.1 From FC-Version 1.10

(1.17 From FC-Version 1.17

(1.19 From FC-Version 1.19

(2.00 From FC-Version 2.00

(2 Analog SINGLE MODE: RES = 1 -> Prepares for a trigger event
STORE MODE: RES = 1 -> Starts the digital acquisition anew.
(Exception SINGLE Mode: Trigger event necessary)

(3 Only in Analog / Digital Scope

(4 The data will be loaded in an EEPROM. This consists of only a limited number ($\geq 1\,000\,000$) of program cycles. Therefore this command should not be used too often unnecessarily.

(5 This function will be reset after a counter has run down (ca. 5s). Each further output before the end of the interval sets the counter anew and thus prolongs the time until reset.

(6 The maximum interval permitted between the transmission of each trace byte is 2s, otherwise the scope will exit the remote state.

(7 Valid for FC-Versions lower 1.05:

For (w w) stands (w,w); comma as ASCII character.

For (w w array) stands (w,w:array); comma and colon as ASCII character.

(8 Only HM1507 and HM1507-2

(9 Only for instruments with 2 timebases.

(10 16 Bit INTEGER value in two's complement based on the graticule center (1000 bit/DIV).

Ex. 1: Set Y position to graticule center Output: 0 dec. = 0 hex.

Ex. 2: Set Y position to +1 division Output: 1000 dec. = 3E8 hex.

Ex. 3: Set Y position to -1 division Output: 64536 dec. = FC18 hex.

Examples

Some examples, with detailed explanations regarding the command, are given below.

Most of the commands are terminated with **CR** (ENTER)= 0Dhex and **LF** = 0Ahex. Also, the scope terminates each acknowledgment string with these characters.

The command parameters (given in brackets) may be ASCII characters (a), or binary values (b).

BELL=(a)

Explanation: This command outputs the tone 2.

String to scope: BELL=2 CR

Character sequence in hex representation: 42 45 4C 4C 3D 32 0D

Answer from scope: 0 CR LF

Answer from scope in hex representation: 30 0D 0A

CH1=(b)

Explanation: This command switches the channel on, with 5mV and AC.
See also byte 1 of DDF.

String to scope: CH1= (52hex) CR

Character sequence in hex representation: 43 48 31 3D 52 0D

Answer from scope: 0 CR LF

Answer from scope in hex representation: 30 0D 0A

ERRBP?

Explanation: This command interrogates the status of the control tone for error. 1 means Error beep is switched on, 0 corresponds to off.

String to scope: ERRBP? CR

Character sequence in hex representation: 45 52 52 42 50 3F 0D

Answer from scope: ERRBP:1

Answer from scope in hex representation: 45 52 52 42 50 3A 31

RDWFM1:(ww)

Explanation : Read signal trace from channel 1.

Parameter 1: Start address in acquisition store = 0 dec. = (00 00)hex.

Parameter 2: Number of bytes to be read = 2048 = (08 00) hex. When transmitting word data, note that the low byte must be output first.
It is possible to read only a part of the acquired signal by transmitting other values for the start address and the number of bytes to read.

String to scope: RDWFM1:(00hex)(00hex)(00hex)(08hex) CR

Character sequence
in hex representation: 52 44 57 46 4D 31 3A 00 00 00 08 0D

Answer from scope
in hex representation: 52 44 57 46 4D 31 3A 08 00 00 08 XX XX ...
... XX ;(2048 Byte XX)

WFMPRE?

String to scope: WFMPRE?

Answer from scope
in hex representation: 57 46 4D 50 52 45 3A XX XX C8 00 19 00 YY YY ZZ ZZ

Explanation:	Byte 1 to 7:	WFMPRE:
	Byte 8 & 9:	Number of the byte at trigger
	Byte 10 & 11:	Resolution in X direction per Div. (200)
	Byte 12 & 13:	Resolution in Y direction per Div. (25)
	Byte 14 & 15:	(YY YY) Y1 position as 16 bit integer variable standardized on 25 per Div., where the value 0 signifies no shift, 25 shifted one division up and -25 shifted one division down.
	Byte 16 & 17:	(ZZ ZZ) Y2 position standardized on 25 per Div.

Calculating the voltage of the sampled signal form:

Given: UN : Voltage value of the Nth sample
25 : Y resolution per Div. (see WFMPRE?)
Y1Pos : Y1 position of the signal form (see WFMPRE? YY YY)
ByteN : Value of the signal form byte (see RDWFM1 XX)
V/Div : Attenuator setting (e.g.: 5mV)

Calculation without taking the Y1 position into account:

$$UN = (\text{ByteN} - 128) / 25 * V/\text{Div}$$

With this method it is only possible to evaluate the voltage difference of the acquired signal, since there is no reference (Zero voltage). In order to calculate the absolute voltage of the sample one should include the Y position in the calculations.

$$UN = (\text{ByteN} - 128 - Y1\text{Pos}) / 25 * V/\text{Div}$$

The deflection coefficient (V/DIV) is obtained from the DDF byte 1 or with the command CH1?.

TRGVAL?

With this command it is possible to evaluate the peak value and the arithmetical mean value of the measured signal. This is shown in the following example.

Scope setting: Channel 1 on; 5mV deflection sensitivity; DC input coupling; probe 1:1; 1kHz calibration; probe in the calibration socket; trigger source - channel 1; AC Trigger coupling; Timebase A set to 200µs; analog mode.

A positive square wave signal with an amplitude of 4 div. can be seen on the oscilloscope display.

String to scope: TRGVAL?

Answer from scope

in hex representation: 54 52 47 56 41 4C 3A D0 07 30 F8 D0 07 XX XX

Explanation: Byte 1 to 7: String TRGVAL:
Byte 8 and 9: Positive peak value (signal)
Byte 10 & 11: Negative peak value (signal)
Byte 12 & 13: Arithmetic mean value (signal)
Byte 14 & 15: Reserved (value undefined)

Positive and negative peak values are integer values referred to the mean value with the weighting of 1000/div.

Then follows: 07D0 hex. = +2000 decimal
F803 hex. = -2000 decimal

Mean value: $U_m = 2000 / 1000 * 5\text{mV} = 10\text{mV}$

positive peak value: $U_{pp+} = 2000 / 1000 * 5\text{mV} + U_m = 20\text{mV}$

negative peak value: $U_{pp-} = -2000 / 1000 * 5\text{mV} + U_m = 0\text{mV}$

Since the trigger amplifier is not calibrated there can be a deviation from the values.

SOURCE FILE: TERM9X_D.TXT /CHEMNITZ /04.DEC.1994

Remote control with the Term9X program of NORTON COMMANDER

1. Settings

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1.1 term90.exe

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Under 'Settings \ Interface':

Port:	Your free serial port, e.g. COM2.
Baud rate:	recommended 19200.
Data bits:	8 BITS
Parity:	None
Stop bits:	2 BITS
Data exchange:	Xon/Xoff to off. RTS/CTS/ to on.

Under 'Settings \ Terminal' mark the 'ANSI'.

Under 'Settings \ ' mark the option 'Echo'.

1.2 term95.exe

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Under 'Settings \ Driver' mark the option 'Standard'.

Under 'Settings \ Line...':

Port:	Your free serial port, e.g. COM2.
Baud rate:	Recommended 19200.
Data bits:	8 BITS
Parity:	None
Stop bits:	2 BITS
Data exchange:	Xon/Xoff to off. RTS/CTS/ to on.

Under 'Settings \ Terminal Emulation' mark the option 'ANSI'.

Under 'Settings \ Terminal settings' mark the option 'Echo'.

2. Remote control

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The transmission can be started after all settings have been correctly set,.

Remote On: Enter one after the other a SPACE and ENTER.
The instrument goes into remote mode, which is evident from the Remote LED.
Now you can enter any desired command such as, for example, 'vers?'.

Example: Switch on channel 2 of the instrument and at the same time set it to 5mV.
Enter 'ch2=', press the key "Alt Gr" and simultaneously enter from the number keyboard one after the other the numbers 0,1 and 8. After releasing the "Alt Gr" key, the instrument will be set.
The RETURNCODE will be displayed behind the transmitted command.

The value 0 ("Alt Gr" 000) will not be transmitted.